



SOC THEORETICAL TASK REPORT

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Introduction

In today's digital landscape, organizations face continuously evolving cybersecurity threats including malware infections, brute-force attacks, phishing campaigns, insider threats, and advanced persistent threats (APTs). To effectively defend against these threats, organizations establish a centralized Security Operations Center (SOC) responsible for continuous monitoring, threat detection, analysis, and incident response.

A Security Operations Center (SOC) acts as the nerve center of an organization's cybersecurity infrastructure. It integrates people, processes, and technology to ensure proactive threat detection, rapid incident response, and minimized business impact. SOC operations are guided by globally recognized frameworks such as the NIST Cybersecurity Framework and MITRE ATT&CK, which provide structured methodologies for identifying, detecting, responding to, and recovering from security incidents.

This report documents the completion of theoretical SOC roadmap tasks, covering core areas such as SOC fundamentals, security monitoring, log management, SIEM operations, incident response lifecycle, and security frameworks. The objective of this report is to demonstrate conceptual understanding of SOC architecture, operational workflow, monitoring tools, and analytical techniques used in modern security operations.

The knowledge gained through this study forms the foundation for practical SOC implementation and hands-on lab activities involving log analysis, alert triage, intrusion detection systems (IDS), and SIEM-based monitoring. This documentation serves as evidence of structured learning and theoretical preparedness for real-world SOC responsibilities.

1. SOC Fundamentals and Operations

1.1 Purpose of a SOC

A Security Operations Center (SOC) is a centralized unit responsible for continuously monitoring, detecting, analyzing, and responding to cybersecurity incidents within an organization.

The primary objectives of a SOC include:

- Proactive threat detection
- Incident response and containment
- Continuous security monitoring (24/7 operations)
- Reducing dwell time of attackers
- Protecting organizational assets and data



A SOC operates using structured processes, defined escalation paths, and standardized security frameworks.

1.2 SOC Roles and Responsibilities

A well-structured SOC includes multiple tiers:

Tier 1 Analyst (L1 – Alert Monitoring)

- Monitor SIEM dashboards
- Review and triage alerts
- Identify false positives
- Escalate suspicious alerts to Tier 2

Tier 2 Analyst (L2 – Incident Responder)

- Perform deeper investigation
- Analyze logs and correlate events
- Identify Indicators of Compromise (IOCs)
- Contain threats if confirmed

Tier 3 Analyst (L3 – Threat Hunter / Expert)

- Advanced malware analysis
- Threat hunting activities
- Detection rule creation
- SIEM tuning and optimization

SOC Manager

- Oversee SOC operations
- Manage team workflow
- Report security metrics (MTTD, MTTR)
- Ensure compliance and governance

Threat Intelligence Analyst

- Monitor global threat feeds
- Integrate IOCs into SIEM
- Map threats to MITRE ATT&CK framework

1.3 SOC Frameworks Studied

NIST Cybersecurity Framework

- Identify
- Protect
- Detect
- Respond
- Recover



MITRE ATT&CK Framework

Used for mapping attacker techniques such as:

- Initial Access
- Privilege Escalation
- Lateral Movement
- Command & Control

2. Security Monitoring Basics

2.1 Objectives of Security Monitoring

Security monitoring ensures:

- Detection of anomalies
- Identification of unauthorized access
- Prevention of policy violations
- Early detection of malware or brute-force attacks

2.2 Tools Studied

SIEM Tools

- Splunk
- Elastic SIEM

Functions:

- Log aggregation
- Correlation rules
- Alert generation
- Dashboard visualization

Network Traffic Analysis

Wireshark

Used for:

- Packet inspection
- Protocol analysis
- Suspicious traffic detection
- ICMP/TCP/UDP monitoring

2.3 Key Security Metrics

- **False Positives**

Alerts incorrectly flagged as malicious.

- **False Negatives**

Malicious activities that were not detected.



- **MTTD (Mean Time to Detect)**

Average time taken to detect a threat.

- **MTTR (Mean Time to Respond)**

Average time taken to respond and contain a threat.

Reducing MTTD and MTTR improves SOC efficiency.

3. Log Management Fundamentals

3.1 Log Lifecycle

The log management process includes:

- Collection – Gathering logs from endpoints, servers, firewalls.
- Normalization – Converting logs into standardized formats (JSON, CEF).
- Storage – Secure retention of logs.
- Retention – Defined storage duration as per compliance.
- Analysis – Querying and correlation for threat detection.

3.2 Types of Logs Analyzed

Windows Event Logs

- Event ID 4625 – Failed login attempts
- Event ID 7045 – New service installation

Used to detect brute-force attacks or persistence techniques.

Syslog

- Network device logs
- Firewall logs

HTTP Logs

- Web server access logs
- Suspicious URL detection
- Status code analysis (404, 500 errors)

3.3 Log Query Practice

Example: Detecting Failed Login Attempts

- Filter for Event ID 4625
- Identify multiple failed attempts from same IP
- Detect brute-force patterns



4. Security Tools Overview

4.1 SIEM – Splunk

- Log ingestion
- Real-time dashboards
- Alert creation
- Correlation rules

4.2 IDS – Snort

- Signature-based detection
- ICMP alert testing
- Custom rule creation

4.3 Vulnerability Scanner – Nessus

- Network vulnerability scanning
- CVE identification
- Risk severity scoring

4.4 Endpoint Monitoring – Osquery

- System query framework
- Endpoint visibility
- Suspicious process detection

5. Basic Security Concepts

5.1 CIA Triad

Confidentiality – Protecting sensitive information

Integrity – Ensuring data accuracy

Availability – Ensuring systems are accessible

5.2 Threat, Vulnerability, Risk

- Threat – Potential attack source
- Vulnerability – Weakness in system
- Risk – Probability of threat exploiting vulnerability

5.3 Defense-in-Depth

Layered security approach:

- Firewall
- IDS/IPS
- Endpoint security
- SIEM monitoring



5.4 Zero Trust Model

- Never trust, always verify
- Continuous authentication
- Least privilege access

6. Security Operations Workflow

- Alert generated from SIEM/IDS
- Tier 1 triage
- Escalation to Tier 2
- Investigation & log correlation
- Containment & eradication
- Documentation
- Lessons learned

7. Incident Response Basics

Based on NIST SP 800-61:

- Preparation
- Identification
- Containment
- Eradication
- Recovery
- Lessons Learned

Each incident must be documented with timeline, root cause, impact, and mitigation steps.

8. Conclusion

All theoretical SOC roadmap tasks were completed successfully.

Concepts including SOC operations, security monitoring, log management, SIEM usage, IDS testing, and incident response lifecycle were studied and documented.

Theoretical understanding has been aligned with practical SOC lab exercises to simulate real-world security monitoring and threat detection scenarios.

References

- NIST SP 800-61 Incident Handling Guide
- MITRE ATT&CK Framework